

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 36

Title: Minerals and nutritional anemias

Dr. Clayton Taylor

Acknowledgement: Dr. Sharmila Upadhya

Email: ctaylor2@sgu.edu

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Objectives: Minerals and nutritional anemias

SOM. MK.III.BPM2.3.DM.3.BCHM.1336	Indicate metabolic roles of trace minerals like iodine, zinc and selenium.
SOM. MK.III.BPM2.3.DM.3.BCHM.1337	Identify role of copper as a cofactor for enzymes (lysyl oxidase)
SOM. MK.III.BPM2.3.DM.3.BCHM.1338	Discuss clinical manifestations, laboratory findings and molecular defects in Wilson disease
SOM. MK.III.BPM2.3.DM.3.BCHM.1339	Review molecular basis of Menkes syndrome.
SOM. MK.III.BPM2.3.DM.3.BCHM.1340	Describe iron metabolism and regulation of its absorption
SOM. MK.III.BPM2.3.DM.3.BCHM.1341	Interpret significance of estimation of serum ferritin, serum iron, transferrin saturation and TIBC in iron deficiency anemia and hereditary hemochromatosis
SOM. MK.III.BPM2.3.DM.3.BCHM.1342	Discuss laboratory findings, inheritance and molecular basis of hereditary Hemochromatosis.
SOM. MK.III.BPM2.3.DM.3.BCHM.1343	Differentiate between nutritional causes of microcytic anemia (Iron, pyridoxine, copper deficiency and lead poisoning) using laboratory findings and clinical presentation.
SOM. MK.III.BPM2.3.DM.3.BCHM.1344	Differentiate between nutritional causes of macrocytic anemia (vitamin B12 and folate) using clinical features and laboratory findings

Recommended Reading

- Lippincott's Biochemistry 9th Edition
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- [Micronutrients: Minerals | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

MINERALS / TRACE ELEMENTS

- Calcium and phosphate (In MSK and ER modules)
- Copper
- Iron
- Zinc
- Fluoride
- Selenium

Mineral	Functions	Deficiency Syndromes
★ Iron	Essential component of hemoglobin and iron-containing metalloenzymes	Hypochromic microcytic anemia
Zinc	Component of enzymes, principally oxidases	Acrodermatitis enteropathica, growth retardation, infertility; delay in wound healing
Iodine	Component of thyroid hormones	Goiter and hypothyroidism
Selenium	Component of glutathione peroxidase : Antioxidant	Myopathy, rarely cardiomyopathy
★ Copper	Component of cytochrome c oxidase, dopamine β -hydroxylase, tyrosinase, lysyl oxidase, and involved in cross-linking collagen and keratin	Muscle weakness, neurologic defects, hypopigmentation, abnormal collagen cross-linking
Fluoride	Mechanism unknown	Dental caries

COPPER

Function	Enzyme/ Protein	Deficiency manifestation
Collagen synthesis	Lysyl oxidase: Forms covalent cross links	Collagen not stable → Increased risk of bleeding/ hemorrhage
Melanin synthesis	Tyrosinase	Light colored hair
Antioxidant (scavenges free radicals)	Superoxide dismutase	
Electron transport chain	Cytochromes	
Copper transport protein	Ceruloplasmin; Also has ferroxidase activity → Mobilizes iron (converts ferrous iron to ferric)	Poor iron mobilization → Iron deficiency anemia

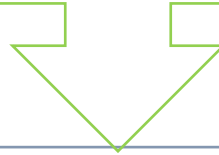
Menkes syndrome (Menkes kinky hair syndrome)

- Inherited defect in dietary copper absorption
- Low plasma copper levels
- X-linked disorder
- Hair twisty, grayish and “kinky” (**Tyrosinase**)
- Aneurysms and neurological dysfunction
(low **lysyl oxidase** activity)
- Early age of presentation (1-2 years)

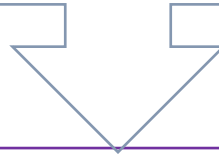


Copper Metabolism

Dietary copper transported to liver



In hepatocytes, forms **ceruloplasmin** (secreted into plasma) (requires **Copper transporting ATPase**)



Aged ceruloplasmin taken up by liver from plasma and **copper secreted into bile** (requires **Copper transporting ATPase**)

Wilson disease

- Autosomal recessive disorder
- Copper excess in **liver, brain and eye (cornea)**
- Mutation of **copper transporting ATPase (*ATP7B* gene)**
 - Over 30 mutations (Allelic heterogeneity)
 - Attaches copper to ceruloplasmin and excretes copper into bile
- Diagnosis:
 - Low serum ceruloplasmin;
 - Copper overload in liver → Altered liver function tests

Defective biliary copper excretion
Defective copper incorporation into ceruloplasmin

Accumulation of toxic levels in liver

Excess copper spills into plasma

Liver damage – hepatitis and cirrhosis (**presenting symptom**)

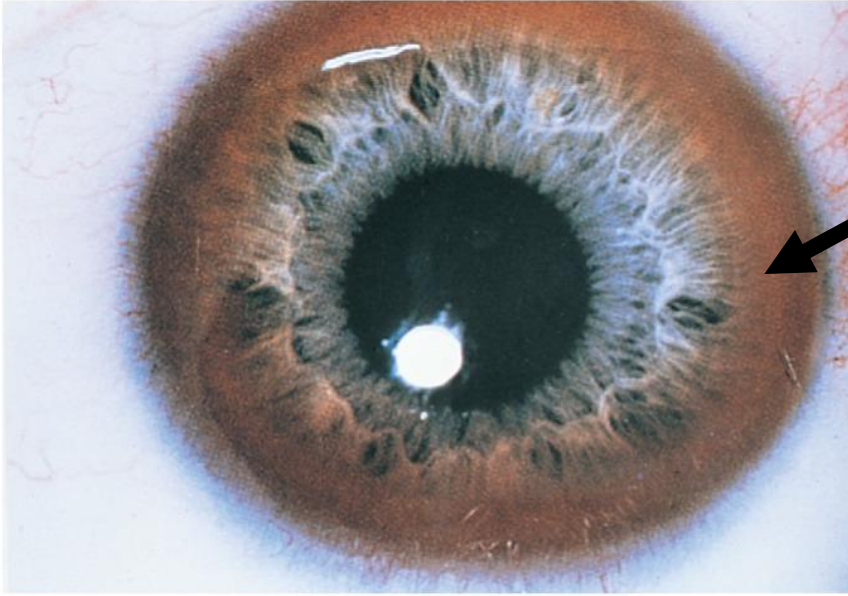
- **Eye:** Kayser-Fleischer rings in cornea
- **Brain:** Neurological damage causing neuropsychiatric symptoms (basal ganglia)

Lab Diagnosis

- Decreased serum ceruloplasmin
- Increased urinary excretion of copper
- Increased hepatic copper content

Wilson disease

Wilson disease: Corneal findings



**Kayser-Fleischer (KF) rings –
Copper deposits in cornea**

Patients with Kayser-Fleischer
rings usually have **basal
ganglia (CNS)** involvement

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VARIABLE	MENKES	WILSON
Whole-body Cu	Low	High
Free serum Cu	Low	High
Urinary Cu	Low	High
Inheritance	X-linked	AR
Cu-transporting ATPase affected	ATP7A	ATP7B

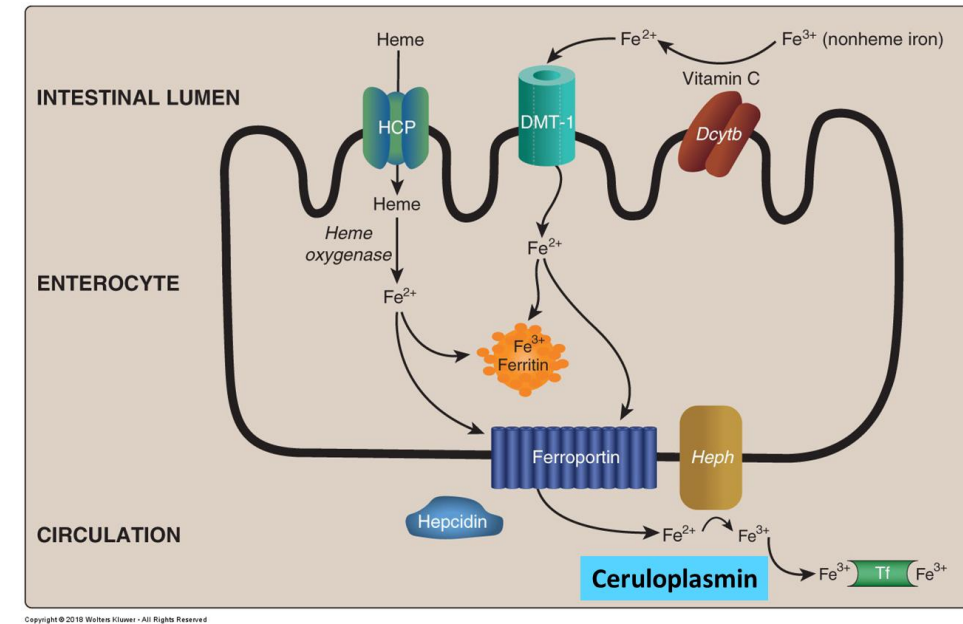
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IRON

- Heme synthesis- **Hemoglobin, myoglobin** (in muscle)
- RBCs contain **hemoglobin** (Largest storage form for **iron**)
 - Loss of RBCs results in loss of iron
- For Redox reactions and component of **cytochromes** of electron transport chain

Dietary iron uptake

- **Tightly regulated** by body iron stores
- Intestinal mucosal cells are iron sensors (requires HFE)
- Higher body iron stores; less iron absorbed in intestine
- **Heme iron (red meat)** absorbed better than inorganic iron (vegetarian)
- **Inorganic iron** in **ferrous form** is absorbed
- Ferric iron changed to ferrous iron in stomach
 - low pH in stomach and dietary vitamin C
- Vegans risk of iron deficiency (high content of phytate and oxalate)



Iron transport and storage

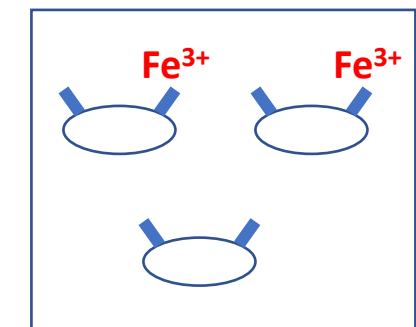
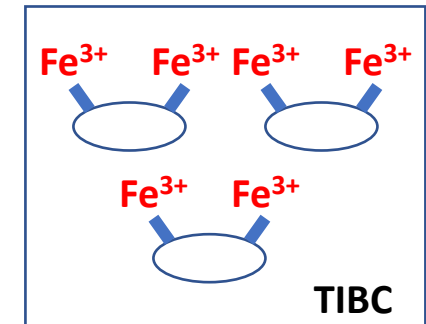
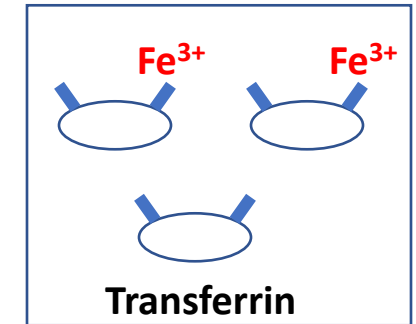
- **Ceruloplasmin** (ferroxidase) converts ferrous iron to **ferric iron** which is incorporated to transferrin
- **Transferrin** (transport protein) for **ferric** ion in plasma
 - β -globulin serum protein
- Iron stored in liver and RES as **ferritin** and **hemosiderin** (ferric)
- Tiny amounts of ferritin in blood – Measurement of **serum ferritin** levels indicators of body iron status

Lab tests for iron status

- Serum ferritin levels indicator of iron stores (Normal: 50-300 ng/ml)
 - High serum ferritin in iron overload (hemochromatosis)
 - Low serum ferritin in iron deficiency anemia
- Serum iron (remember, iron is bound to transferrin)
 - Low serum iron levels in iron deficiency anemia
 - High serum iron levels in iron overload (hemochromatosis)
- Transferrin saturation and Total iron binding capacity
- Hemoglobin levels and hematocrit
- MCV, MCHC – Red cell indicators of RBC size and hemoglobin concentration

Lab tests for iron status

- **Serum iron:** Iron bound to transferrin (Normal: 50-150 $\mu\text{g/dL}$)
 - Serum iron decreased in iron deficiency anemia
 - Serum iron increased in iron overload
- **Total iron binding capacity (TIBC):** Total number of iron binding sites on transferrin (transferrin levels) Normal: 250-400 $\mu\text{g/dl}$
 - TIBC increased in iron deficiency anemia
 - TIBC decreased in iron overload
- **Percent saturation of transferrin:** Percent of transferrin binding sites occupied by iron. Normally $\frac{1}{3}$ binding sites saturated with iron (Normal: 30% transferrin saturation) [Total Iron Binding Capacity \(TIBC\) Explained](#)
 - Transferrin saturation decreased in iron deficiency anemia
 - Transferrin saturation increased in iron overload



Iron deficiency anemia

- Most common nutritional deficiency
- Reduced dietary iron absorption: Vegans, low dietary vitamin C, achlorhydria (low stomach HCl), high dietary fiber
- Infants, pregnant women and blood donors at risk
- **Chronic bleeding:** Increased menstrual bleeding, uterine fibroids, GI bleeding (ulcer/ cancer), parasitic infestations

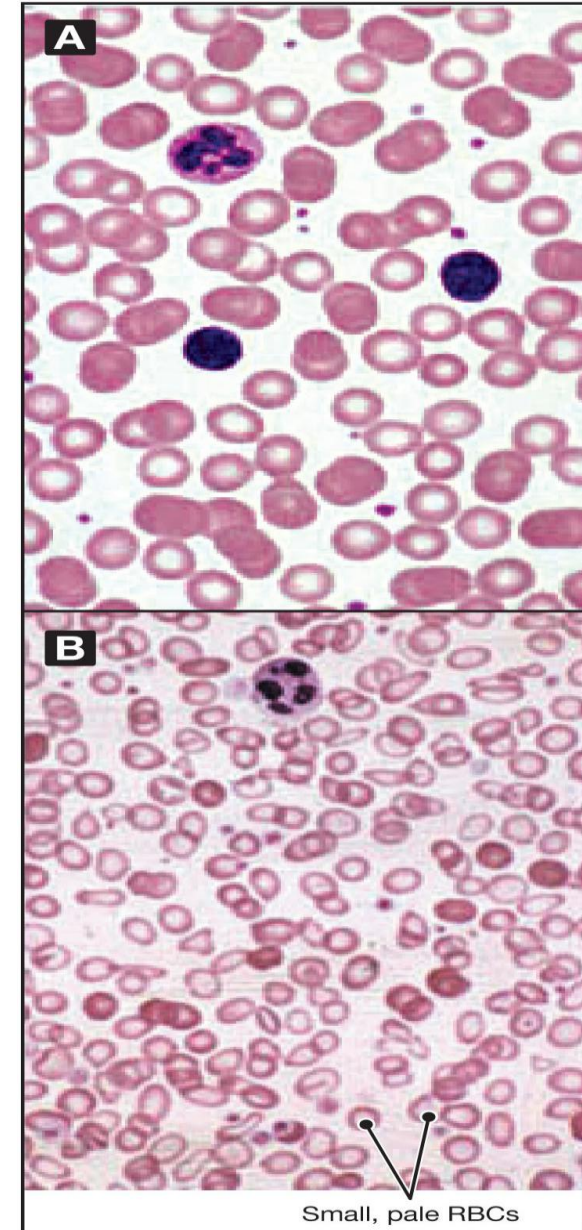
Clinical features:

- Iron deficiency anemia – hypochromic microcytic
- Fatigue and pallor
- Weakness, dizziness
- Brittle nails
- Pica (appetite for soil)



Iron deficiency anemia: Lab tests

- Low serum ferritin levels (indicate low body iron stores)
- Low serum iron levels
- Low percent saturation of transferrin (<20%)
- Increased TIBC
- Low hemoglobin levels; low hematocrit
- Low MCV (microcytic anemia) and low MCHC (hypochromic anemia)
- **Bone marrow iron stores - low**



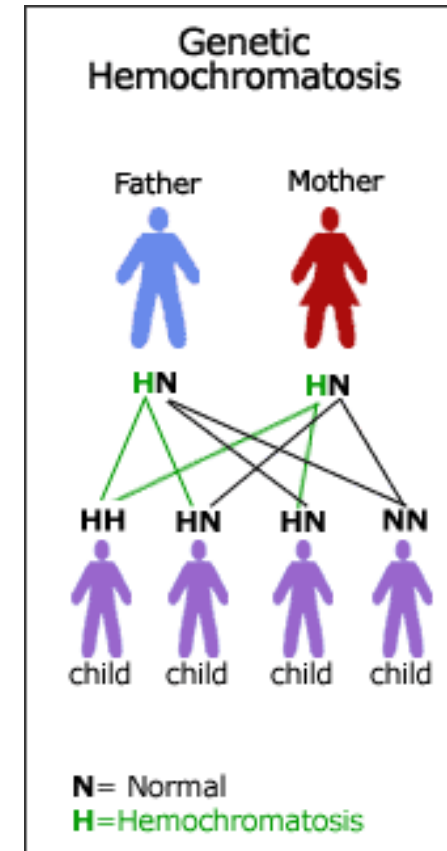
Dietary modifications for management of iron deficiency



- Supplement iron rich foods
- Increase availability of dietary iron – vitamin C; Include heme iron (animal sources); Reduce dietary fiber

Hereditary Hemochromatosis (Iron overload)

- Autosomal recessive disorder; Common in Caucasians
- **Excessive iron absorption** (*HFE* gene mutation)
- Most frequent mutation is **C282Y allele** of HFE
 - However, other mutations are known (**allelic heterogeneity**)
 - Many are **compound heterozygotes**
- Delayed age of onset
- Males affected earlier than females – variable expressivity
- Excessive accumulation of iron in **skin, liver and pancreas**



Hereditary Hemochromatosis (Iron overload)

- Excess iron causes free radical formation and DNA damage

Clinical Features: <http://www.hemochromatosis.org/#videos>

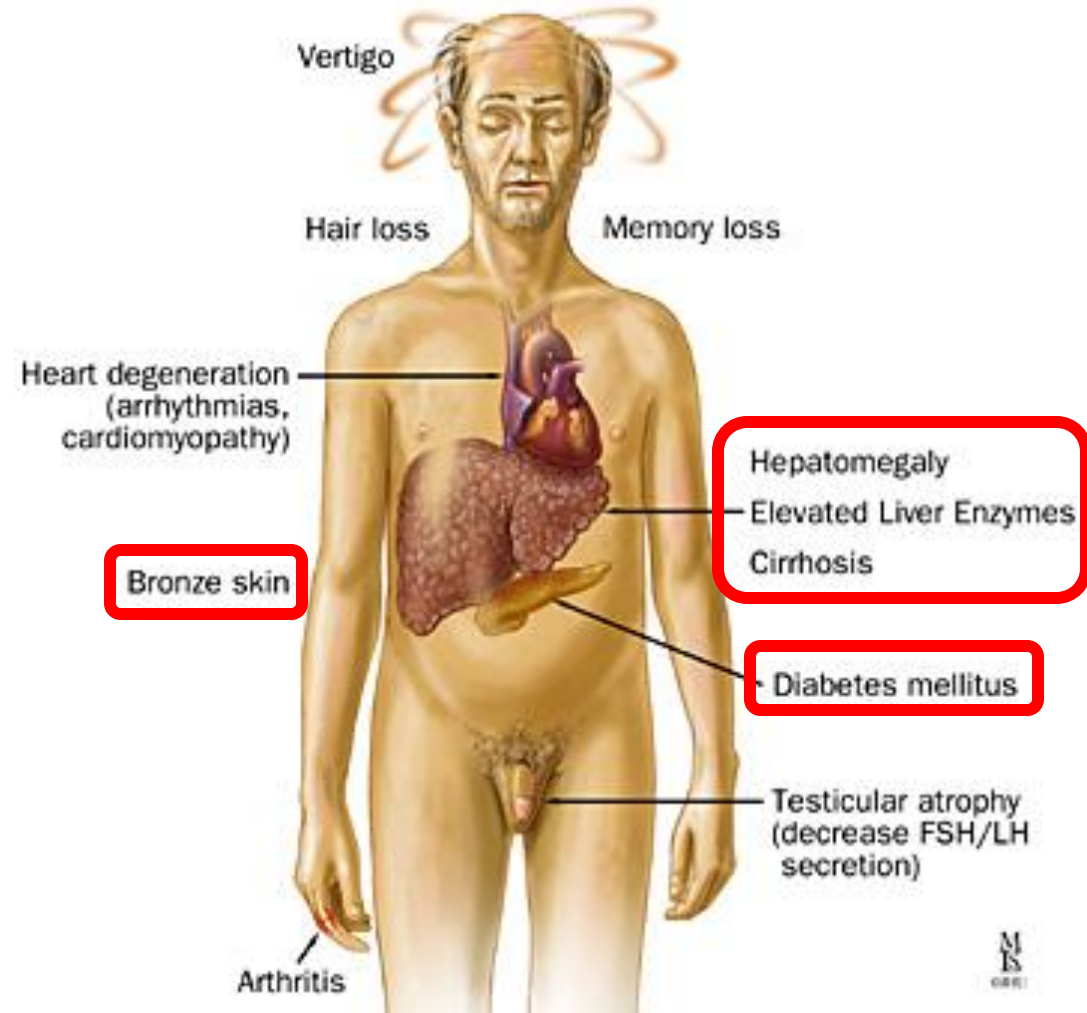
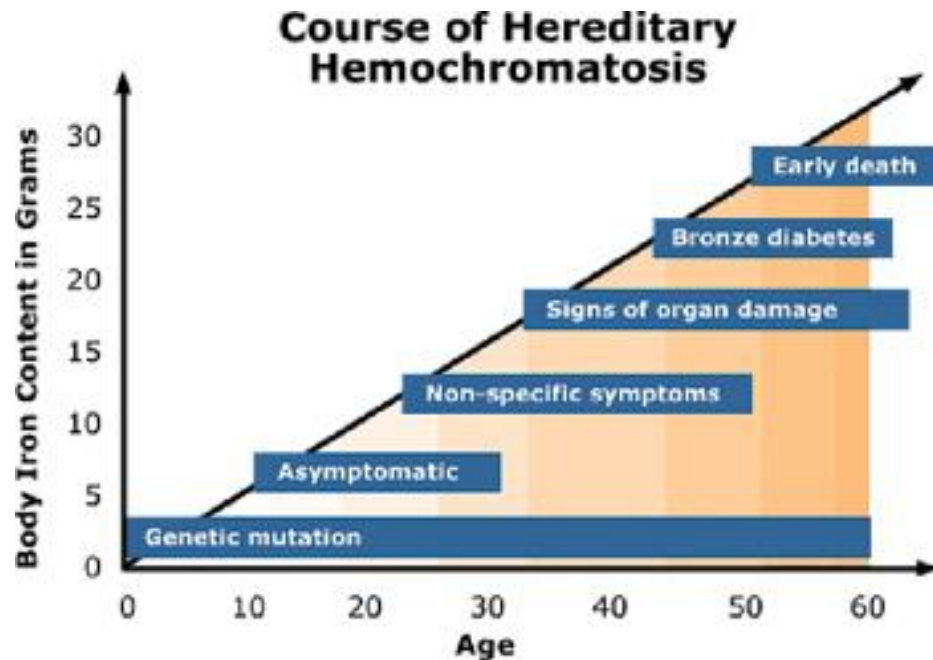
- More common in males; presenting age 40
- Acute synovitis or chronic joint pain (knuckles)
- Chronic fatigue (difficult to diagnose)
- ★• Hepatomegaly; Liver damage – cirrhosis; hepatocellular carcinoma
- ★• Diabetes – destruction of beta cells of pancreas
 - Cardiac dysfunction
- ★• Brownish skin pigmentation (Bronze orange color of skin)
 - ‘**Bronze Diabetes**’



Figure 1 – The hand on the right is that of a 60-year-old white man with hereditary hemochromatosis. It is shown next to the hand of an unaffected person for comparison. The hyperpigmentation—characteristically golden brown or bronze—is generalized, but appears most prominently on sun-exposed or traumatized areas.
(Courtesy of Eugene Wong, MD, and Peter Panopoulas, MD.)

Manifestations of hemochromatosis

<http://www.spacecoastmedicine.com/2011/01/hemochromatosis-is-most-common-disease-you%E2%80%99ve-never-heard-of.html>



Hereditary Hemochromatosis – Lab tests

- High serum ferritin levels – indicate excessive iron stores
- High serum iron levels
- Transferrin saturation (>50%)
- TIBC is decreased
- Altered liver function tests: Iron overload in liver

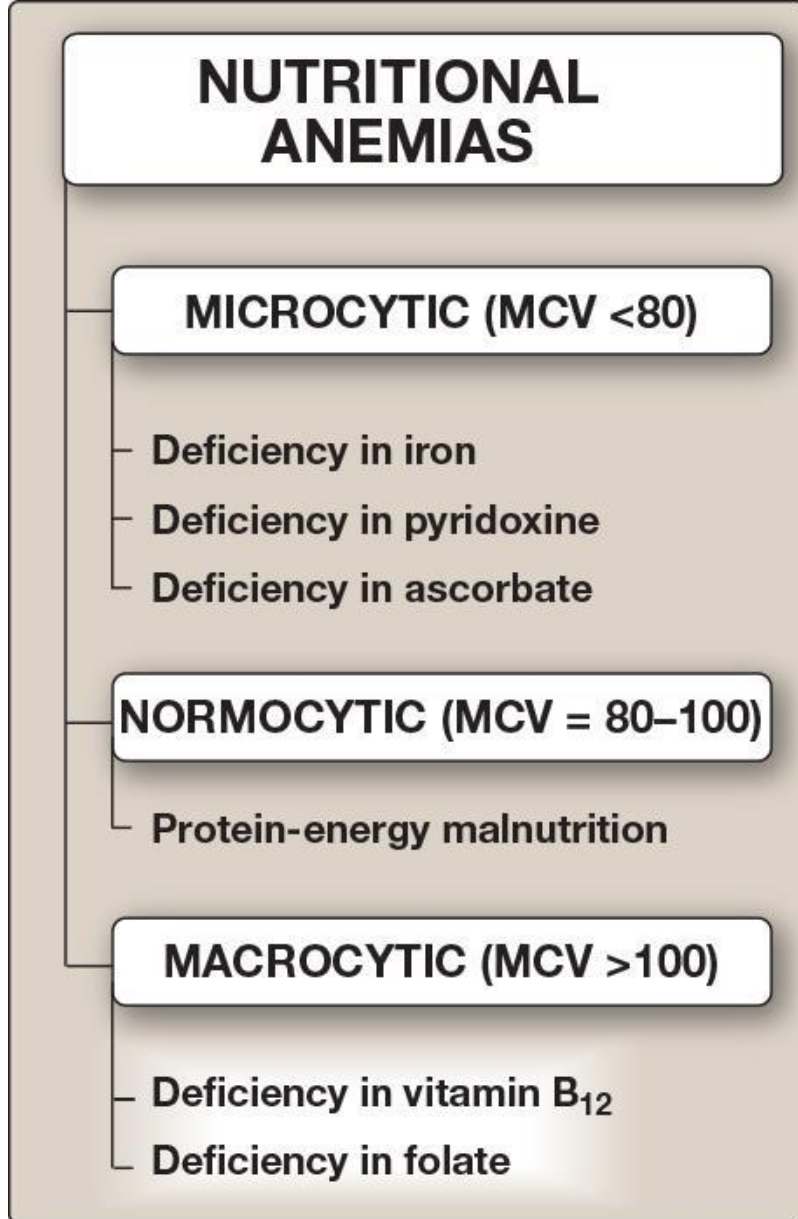
Management

- Phlebotomy – removal of blood (donation of blood)
- Dietary modifications – Reduce dietary iron content and iron absorption from diet
 - Reduce red meat (iron in meat is better absorbed)
 - High fiber content (fiber reduces iron absorption)

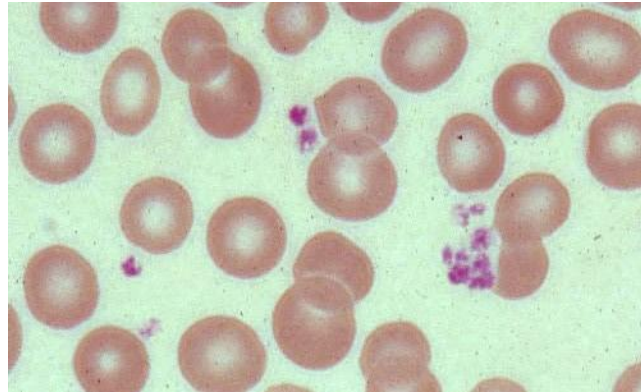
Nutritional anemia

- Anemia is a clinical sign
- Reduced red cell mass
- Hemoglobin levels <12 g/dL; and/ decreased RBC count
- Measure hemoglobin concentration, hematocrit and RBC count
- Erythropoiesis (RBC formation) in bone marrow depends on **availability of nutrients**, growth factors and precursor cells
- Reduced nutrient availability results in nutritional anemia

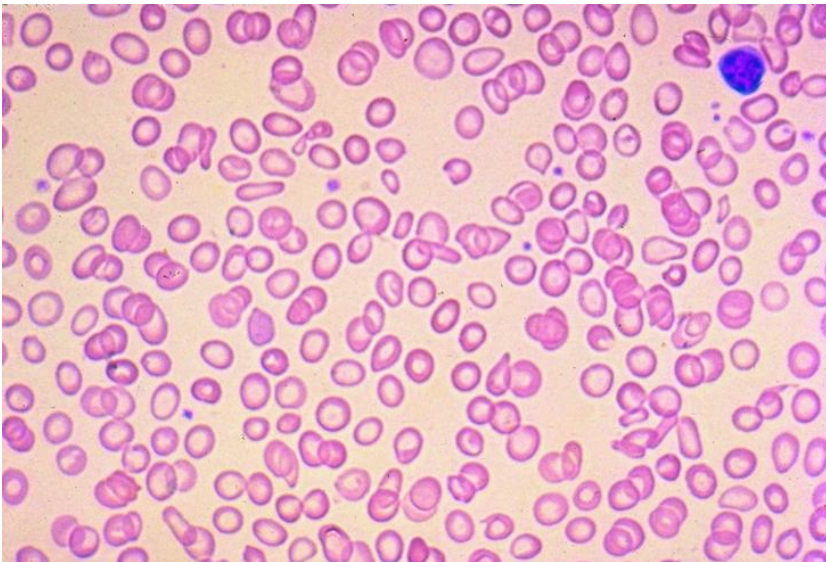
Nutritional anemia: Classification based on MCV



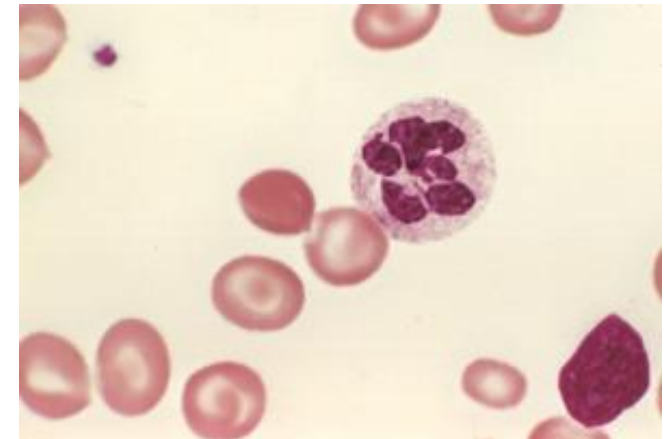
- Nutritional anemias based on RBC size
 - **Microcytic anemia** (MCV<80) due to reduced heme synthesis.
 - Also seen in lead poisoning
 - **Macrocytic anemia** (MCV>100) due to reduced cell division.
 - **Normocytic anemia** (Normal MCV) – Protein energy malnutrition (PEM)



Normal



**Microcytic,
hypochromic anemia**



**Macrocytic,
hypersegmented
neutrophils**

Types of anemia and lab tests in nutritional anemia:

	Type of anemia	Lab tests
Iron deficiency	Microcytic hypochromic anemia	Low MCV, Low MCHC; Low serum ferritin; Low serum iron; Increased TIBC; Decreased %Transferrin saturation; Low Bone marrow iron stores
Pyridoxine deficiency	Microcytic anemia	Low MCV; Elevated Homocysteine; Iron parameters - Normal
Ascorbate (Vitamin C) deficiency	Microcytic anemia	Low MCV; Scurvy (Characterized by sore and spongy gums, loose teeth, fragile blood vessels, hemorrhage, swollen joints, bone changes, and fatigue)
Vitamin B12 deficiency	Macrocytic, megaloblastic anemia; Neuropsychiatric manifestations	MCV – Increased; Increased Homocysteine; Increased methylmalonate; Low Vitamin B12 levels
Folic acid deficiency	Macrocytic, megaloblastic anemia	MCV – Increased; Increased Homocysteine; Methylmalonate - Normal; Low Folic acid levels; Normal vitamin B12 levels

Comparison of macrocytic anemias

Vitamin B12 deficiency

- Macrocytic anemia with associated **neurological features**
- Anemia – Macrocytic (MCV increased) and megaloblastic bone marrow
- Serum **homocysteine** and serum **methylmalonate levels are elevated**
- Macrocytic anemia due to folate trap
- **Neurological features due to accumulation of methylmalonate (affects myelination)**
- Risk factors: Parietal cell destruction (pernicious anemia); ileal mucosal disease or resection; vegan diet
- Serum vitamin B12 levels are low
- Significant B12 liver stores

Folate deficiency

- Macrocytic anemia
- Anemia – Macrocytic (MCV increased) and megaloblastic bone marrow
- Elevated serum homocysteine levels; **Normal serum methylmalonate levels**
- Macrocytic anemia due to folate deficiency
- **NO neurological features**
- Risk factors: pregnancy, chronic alcoholics, folate antagonist treatment
- Serum folate levels are low
- No significant folate stores; Deficiency develops early

Vitamin/ mineral	Biochemical function	Deficiency/ toxicity manifestations		Lab tests
Vitamin A	11 cis retinal – rhodopsin- dim light vision	Night blindness, bitot spots, xerophthalmia, keratomalacia	Hypervitaminosis A: Mimic brain tumors; benign intracranial hypertension	
Vitamin D	1,25 DHCC (calcitriol) – increases serum calcium and phosphate absorption in intestine;	Rickets ; osteomalacia	Hypervitaminosis D – Increased calcification in soft tissues	Low serum calcium and phosphate; Low 25-OH cholecalciferol
Vitamin K	Glutamate gamma carboxylase for gamma carboxylation of factors II, VII, IX and X	Increased prothrombin time; increased bleeding tendency		Prolonged prothrombin time
Thiamine	PDH complex, alpha keto glutarate DH complex; branched chain alpha keto acid DH complex; transketolase	Beri-beri – wet and dry forms; Wernicke-Korsakoff syndrome		Low transketolase activity
Riboflavin	Succinate DH; Acyl CoA DH; PDH complex	Glossitis; cheilosis		
Niacin	NAD and NADP required for DH and reductases NAD and NADP formed from tryptophan	Dermatitis, dementia, diarrhea, death; Carcinoid syndrome and Hartnups disease have high risk of NAD deficiency		
Pyridoxine	Aminotransferases (ALT and AST); Amino acid decarboxylase; delta ALA synthase; Cystathionine synthase (Homocysteine metabolism)	Neurological manifestations; microcytic anemia		
Vitamin B12	MethylmalonylCoA mutase; Methionine synthase	Macrocytic anemia due to folate trap; Neurological manifestations due to accumulation of methylmalonate		Elevated methylmalonate and elevated homocysteine
Folic acid	Homocysteine methyl transferase (methionine synthase); Purine and thymidine synthesis (DNA synthesis (cell division)	Macrocytic anemia due to less purine and thymidine synthesis		Elevated homocysteine levels; Normal methylmalonate levels
Biotin	Propionyl CoA carboxylase; Acetyl CoA carboxylase; Pyruvate carboxylase	Dermatitis		Elevated propionate levels
Vitamin C	Prolyl hydroxylase and lysyl hydroxylase	Collagen stability and formation of H-bonds in collagen		
Copper	Lysyl oxidase (collagen cross links); tyrosinase (melanin synthesis)	Menkes: decreased copper absorption	Wilson: Copper overload and stored in liver, basal ganglia and cornea	
Iron	Heme synthesis and hemoglobin synthesis	Iron deficiency anemia		Low serum iron; Low ferritin; Increased TIBC
		Hemochromatosis: Iron overload; Iron stored in liver, skin and pancreas (bronze diabetes)		High serum iron; High ferritin; Low TIBC